

Mixed Reality Support for Bridge Inspectors

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ABSTRACT

Bridge inspectors work for the safety of our infrastructure and mobility. In regular intervals, they conduct structural inspections – a manual task with a long-lasting and firmly normed analogue tradition. We propose a mixed analogue and digital workflow that includes Mixed Reality views that can be ready-to-hand for bridge inspectors during their work at and in a bridge. Our presented demonstrator was iteratively designed in a collaborative research project and turned into a tablet-based application to digitally support that work. It employs BIM data that contains 3D geometry-data and additional data about the structure, such as previous damage reports.

Keywords: Mixed Reality, Building Information Modeling, Bridge Inspection, Digitalization, Workflow.

Index Terms: •Human-centered computing~Collaborative and social computing~Collaborative and social computing systems and tools •Information systems~Information systems applications~Decision support systems~Expert systems

1 INTRODUCTION

Roads and bridges are crucial parts of our transport infrastructure and thus essential for our mobility at large. Regular structural inspections are needed to ensure their safety. In Germany, there are 39,500 bridges in the net of national highways alone. Further bridges are in the responsibility of federal states and local authorities. Bridge inspectors are working collaboratively on their examination – a manual task with lots of hammering, currently widely documented using pen, paper, and digital photos on-site and a PC-based software for structured entry in the office.

While we do not claim to digitalize the inspection process itself, we believe that the documentation process on-site can benefit from digital tools, including Mixed Reality (MR) views. For the inspection itself, the manual character combined with the knowledge and tradition of the bridge inspector is inevitable to identify all possible damages that may occur. To achieve that support for bridge inspectors, we developed a prototypical MR application demonstrator that employs Building Information Modeling (BIM) data. BIM is a method to accompany the life-cycle of buildings and structures [1]. This method is mandatory for new public construction projects in several countries across the world. But it is also getting increasingly adopted in the private sector. The main advantage of BIM is the digital data exchange

between the parties involved throughout the process and earlier detection of conflicts before construction, resulting in better cooperation, overview, and cost reduction. BIM offers advantages in the maintenance of buildings: The as-built-BIM-model contains every information of the structure, such as the current condition (damages, structural changes, etc.). We report on a MR prototype that was iteratively built with practitioners over 18 months to digitally support bridge inspectors using BIM data on site.

2 RELATED WORK

Characteristically for BIM is one central digital model that contains information on the geometry of a building and additional data like employed materials or further information on producers etc. Regarding the data exchange, file formats play a significant role. The non-profit buildingSMART organization addresses standardization of such file formats. For example, they standardized the Industry Foundation Classes (IFC) file format available for data exchange, IFC-Bridge as extension for bridges, and the BIM Collaboration Format (BCF).

Related systems have been presented, such as the early “Tinmith-Metro” system for mobile modelling of outdoor structures [2], or the more recent “Corsican twin”, a system that employs available BIM and CAD models for mobile use [3]. Garbett et al. built a collaborative multi-user BIM-AR system to support the design and construction using marker-based tracking [4]. Hansen and Kjems focus on the interaction challenges with infrastructure data in the field, demonstrating prototypes for a bridge and a highway [5]. Industrial applications visualizing and interacting with BIM data include products from companies such as Trimble, HoloBuilder, vGIS, or GammaAR.

Combining Computer Supported Cooperative Work (CSCW) and MR can be beneficial, as pointed out by Ens et al. As they state, MR can benefit from the CSCW knowledge, and CSCW concepts – like role management – can be adopted and applied for MR applications [6]. The structural inspection working area has a long-lasting tradition with several rules, regulations, and norms. In Germany, the “Deutsche Industrie Norm” DIN 1076 norm contains guidelines for bridge inspection performances and covers the required documents for monitoring and examination. Besides that, other laws and regulations apply, depending on the bridge’s ownership.

3 SYSTEM DESIGN

The project developed a MR and a VR prototype. This poster focusses on the MR prototype that is mainly meant to be used on-site, i.e. at and in the bridge. The system design is similar for both prototypes, though. We selected Unity as a technical basis, as it provides us with support for desktop, HMD, and mobile environments. Additionally, we use the integrated Unity ARFoundation library and the underlying ARKit for realizing our MR prototype.

We chose an iPad Pro for the MR system. Another option would have been to use smart glasses, like the Microsoft HoloLens. However, the expert bridge inspectors preferred the tablet option after comparing both in a kick-off workshop. Their

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Figure 1: MR-view with half-transparent model overlay



Figure 2: Damage recording with integrated measurement tool

main reasons were the devices' form factor, the more intuitive handling of the tablet (they already used ruggedized Windows tablets in their work), the HoloLens not being practical for being used in a bridge (limited field of view, gesture input system), and the cost. A tablet, in contrast, allows for shared experiences and thus supports collaboration, as other studies show [7], [8].

We first conducted expert interviews with nine field experts to identify their requirements and to define the process of a digitally supported bridge inspection. Then, we tested our initial prototypes with bridge inspectors from our project team in several workshops on site at a nearby test bridge. The bridge inspectors' comments and the observed technical challenges and difficulties fed back into the design and development process, leading to an inherently iterative approach. We pursued an indirect workflow for the model inclusion, converting and optimizing the original IFC file to FBX before importing it into our application splitting into three scenario views. The data is stored in a common data environment (CDE) and synchronized on the tablet through a REST interface. We use the BSCW (Basic Support for Collaborative Work) groupware system as the public administration for federal highways in Germany already utilizes that software – although we require a newer version. To localize the tablet and link the digital model to the real world, we apply marker tracking and the iPad's SLAM capabilities. We use the latter also for a measurement tool (cf. Figure 2). The user can change the opacity of the virtual model using a slider and even select to disable it entirely. He can add a damage entry by pressing a button (top right, Figure 1). The tablet then automatically takes a picture and records the current pose of the camera. He may then enter the details of the damage using a form that reflects exemplarily the logic of the current state-of-the-art PC-based tool "SIB-Bauwerke" (cf. Figure 2).

4 CONCLUSION AND OUTLOOK

From the beginning of the developments, bridge inspectors were involved with their requirements for a support tool to ensure that their needs were considered. This is essential, especially in the case of bridge inspection with its long tradition, to bring developments from research into practice in the future. It will also be crucial for further research to first demonstrate concepts and new processes on a pilot basis, as was done here, to enable tests

and feedback from potential users. We will present a detailed evaluation of our system in a future publication.

Regarding current and future research in this area, it will be particularly relevant to track the rapid development of digital technologies to identify potential for optimized lifecycle management of bridge structures at an early stage and to harmonize it with best practices. Further support possibilities are seen, e. g. using artificial intelligence. However, the bridge inspector, with their specialist experience and knowledge, always plays a central role in this. Thus, the combination of CSCW and MR (as already described by [6]) for BIM seems promising to us. In addition, the high safety standard of bridge inspection according to existing regulations must be maintained while incorporating new insights from research into the regulations to allow for digitally augmented workflows finding their way into practice. Similarly, public tenders and purchasing processes for soft- and hardware need to reflect the usage of MR and VR, amongst others. Finally, we underline that collaborative information systems design, esp. for outside use and practical deployment, should not only be done in an iterative, user-centered way, but also consider organizational, social and physical dimensions for embodied interaction [9], [10].

ACKNOWLEDGEMENTS

This work is based on parts of the research project carried out at the request of the German Federal Ministry for Digital and Transport, represented by the Federal Highway Research Institute, research project No. 15.0666/2019/LRB including its final report „Structural inspections using 3D building models and augmented/virtual reality“. The authors are solely responsible for the content.

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